

Exact Tightness on the Second Subset Diagonal of the Avoid-or-Compress Lemma

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Abstract

Let $\mathcal{A} = (Q, \Sigma, \delta)$ be a synchronizing deterministic finite automaton, let $\emptyset \neq A \subsetneq S \subseteq Q$, and call a word w successful if either $A \not\subseteq Sw$ or $|Sw| < |S|$. The avoid-or-compress lemma gives a successful word of length at most $n - |A|$, where $n = |Q|$. We determine the exact worst-case value on the second subset diagonal $|S| - |A| = 2$. If $S = Q$, the exact value is 1. If $S \subsetneq Q$, the bound $n - |A| = n - |S| + 2$ is sharp, even after simultaneously requiring a binary alphabet, strong connectivity, and synchronization. We give explicit extremal families for every proper-subset parameter pair and prove exactness by a safe-subset invariant. Strong connectivity is immediate from the construction, while synchronization follows from a cycle-and-adjacent-merge argument. An independent breadth-first verifier checks 1,140 parameter pairs with $1 \leq |A| \leq 30$ and $3 \leq n - |A| \leq 40$, with no failures. Thus the general avoid-or-compress upper bound is exactly tight along the entire proper second diagonal under the binary strongly connected synchronizing restrictions.

1 Introduction

Avoiding and compressing words are two of the basic local mechanisms used in the theory of synchronizing automata. Szykuła’s avoid-or-compress lemma states that, whenever a proper target subset $A \subsetneq S$ can eventually be avoided from the image of S , one can either avoid A or compress S within $n - |A|$ steps [1, 2]. The recent open-problems survey emphasizes that tight parameter-dependent bounds are largely unknown, especially when one insists that extremal witnesses remain synchronizing and strongly connected [2].

This paper closes the first nontrivial proper diagonal beyond the fixed-state case. We impose

$$|S| - |A| = 2$$

and determine the exact worst-case length. The answer is particularly clean: except at the full-set boundary $S = Q$, the general upper bound $n - |A|$ is attained exactly by an explicit binary, strongly connected, synchronizing family.

The constructions are simple enough that the lower bound can be proved by tracking only the two states of $S \setminus A$. The parity of $L = n - |A|$ determines whether the extremal word is $b^{L-1}a$ or $b^{L-2}a^2$. The synchronization proof is independent of the avoid-or-compress lower bound: after a short entry word the full state set lies in a core on which one letter acts as a full cycle, while a second word induces one adjacent merge. Conjugating that merge around the cycle yields a reset word.

2 Preliminaries

We write the action of a word on the right. Thus qw is the state reached from q after reading w , and for $X \subseteq Q$,

$$Xw = \{qw : q \in X\}.$$

Definition 1 (Avoid-or-compress distance). Let $\emptyset \neq A \subsetneq S \subseteq Q$. Define

$$\tau_{\text{ac}}(\mathcal{A}; S, A) = \min\{|w| : A \not\subseteq Sw \text{ or } |Sw| < |S|\}.$$

A word counted by τ_{ac} is called *successful*; all shorter words are *safe*. The avoid-or-compress lemma gives the upper bound

$$\tau_{\text{ac}}(\mathcal{A}; S, A) \leq n - |A| \tag{1}$$

whenever the hypotheses of the lemma hold; in particular this applies in the synchronizing setting considered here [1, 2].

For the second diagonal write

$$r = |A|, \quad |S| = r + 2, \quad L = n - r = n - |A|.$$

For a proper subset $S \subsetneq Q$ we have $L \geq 3$.

3 Main theorem

Theorem 1 (Exact second diagonal). *Fix integers n and s with $3 \leq s \leq n$, and let $|A| = s - 2$.*

(i) *If $s = n$, then every synchronizing n -state DFA satisfies*

$$\tau_{\text{ac}}(\mathcal{A}; Q, A) = 1,$$

and this value is exact.

(ii) *If $s < n$, then every synchronizing n -state DFA satisfies*

$$\tau_{\text{ac}}(\mathcal{A}; S, A) \leq n - s + 2,$$

and for every such parameter pair there exists a binary, strongly connected, synchronizing automaton with

$$\tau_{\text{ac}}(\mathcal{A}; S, A) = n - s + 2.$$

Consequently the general upper bound $n - |A|$ is exactly tight on the entire proper second diagonal, even under the binary, strongly connected, and synchronizing restrictions.

The upper bound is (1). The rest of the paper constructs and verifies the matching lower bound.

4 Extremal construction

Fix $r \geq 1$ and $L \geq 3$. Let

$$A = \{A_0, \dots, A_{r-1}\}, \quad B = \{B_0, \dots, B_{L-1}\},$$

$$Q = A \cup B, \quad S = A \cup \{B_0, B_1\}.$$

The alphabet is $\{a, b\}$. Letter b fixes A and cyclically permutes B :

$$A_i b = A_i, \quad B_j b = B_{j+1 \bmod L}.$$

On A , letter a is the chain

$$A_0 \rightarrow A_1 \rightarrow \cdots \rightarrow A_{r-1} \rightarrow B_0.$$

Its action on B depends on the parity of L .

If L is odd, set

$$B_j a = \begin{cases} A_0, & j \text{ even}, \\ B_1, & j \text{ odd}. \end{cases} \quad (2)$$

If L is even, set

$$B_j a = \begin{cases} A_0, & j \text{ even}, \\ B_1, & j \text{ odd and } j < L-1, \\ B_2, & j = L-1. \end{cases} \quad (3)$$

5 Exact avoid-or-compress distance

Lemma 1 (Odd L safe corridor). *Assume L is odd. Every word of length less than L is safe, while*

$$w = b^{L-1}a$$

is successful. Hence $\tau_{\text{ac}}(\mathcal{A}; S, A) = L$.

Proof. For $0 \leq k \leq L-1$ define

$$S_k = A \cup \{B_k, B_{k+1 \bmod L}\}.$$

We have $Sb^k = S_k$. If $0 \leq k \leq L-2$, then k and $k+1$ have opposite parity. By (2), the two B -states are sent to $\{A_0, B_1\}$, while the A -chain contributes

$$\{A_1, \dots, A_{r-1}, B_0\}.$$

Thus $S_k a = S$ for $k \leq L-2$.

Every word can be written as alternating blocks of b 's separated by occurrences of a . Consider any occurrence of a reached from S after a block b^k with $0 \leq k \leq L-2$. The identity $S_k a = S$ shows that the whole block $b^k a$ returns the active subset exactly to S . Hence, in a shortest successful word, every such regular-corridor block may be deleted without changing the subset from which the remaining suffix starts. Repeating this deletion removes every occurrence of a before the wrap-around defect. Thus any shortest successful word must begin with at least $L-1$ consecutive copies of b before its first effective a , so no word of length less than L can succeed. After exactly $L-1$ applications of b , the active pair is $\{B_{L-1}, B_0\}$. Since L is odd, both indices are even, so a maps both states to A_0 . The image loses one state and is compressed. Thus $b^{L-1}a$ is successful of length L . $\square$

Lemma 2 (Even L safe corridor). *Assume L is even. Every word of length less than L is safe, while*

$$w = b^{L-2}a^2$$

is successful. Hence $\tau_{\text{ac}}(\mathcal{A}; S, A) = L$.

Proof. For $0 \leq k \leq L-3$, the consecutive pair $\{B_k, B_{k+1}\}$ contains one even state and one ordinary odd state. Equation (3) therefore gives $S_k a = S$, exactly as in the odd case. Thus every occurrence of a before the final two-state defect simply returns the active subset to S .

After $L-2$ applications of b , the active pair is $\{B_{L-2}, B_{L-1}\}$. The first a maps it to $\{A_0, B_2\}$, so

$$Sb^{L-2}a = A \cup \{B_0, B_2\}.$$

This image still contains all of A and still has size $r + 2$, hence is safe. A second a sends both B_0 and B_2 to A_0 , causing compression. Therefore $b^{L-2}a^2$ is successful and has length L .

As in the odd case, write a word as alternating blocks of b 's separated by occurrences of a . Whenever an a is applied after b^k with $0 \leq k \leq L - 3$, the identity $S_k a = S$ shows that the block $b^k a$ returns exactly to S . Such a regular-corridor block can therefore be deleted from any shortest successful word. After all such deletions, a shortest candidate must first reach the defect directly from S , which requires $L - 2$ consecutive copies of b . The next letter a occurs at total length $L - 1$ and gives $A \cup \{B_0, B_2\}$, which is still safe. Consequently every successful word needs at least one further letter and hence has length at least L . Since $b^{L-2}a^2$ succeeds at length L , no shorter successful word exists. $\square$

Lemmas 1 and 2 prove the lower-bound part of Theorem 1 for every proper subset.

6 Strong connectivity and synchronization

Proposition 1 (Strong connectivity). *Every automaton in the construction is strongly connected.*

Proof. From every A_i , repeated applications of a reach B_0 . Letter b then reaches every B_j . From any even-indexed B_j , letter a reaches A_0 . Hence every state reaches A_0 , and A_0 reaches every state. $\square$

For synchronization define the core

$$C = A \cup \{B_0\}, \quad m = |C| = r + 1.$$

On C , letter a is the m -cycle

$$A_0 \rightarrow A_1 \rightarrow \cdots \rightarrow A_{r-1} \rightarrow B_0 \rightarrow A_0.$$

Write $\alpha = a|_C$.

Lemma 3 (Entry into the core). *If L is odd, the word*

$$w_0 = ab^{L-1}a$$

maps Q into C . If L is even, the word

$$w_0 = a^2 b^{L-2} a^2$$

maps Q into C .

Proof. This follows by direct substitution from (2) and (3). The first one or two a 's collapse all B -states into a constant-size set, the power of b positions that set at the unique wrap defect, and the final a or a^2 places every image in $A \cup \{B_0\}$. $\square$

Lemma 4 (Adjacent merge on the core). *There is a word γ whose restriction to C fixes every core state except one state, which is mapped to its predecessor on the α -cycle.*

Proof. For odd L set

$$u = bab^{L-1}a,$$

and for even L set

$$u = b^{L-1}a^2.$$

A direct evaluation on C shows that u acts as α^2 except that the final two positions of the α -cycle have the same image. Therefore

$$\gamma = u\alpha^{m-2}$$

is the identity on all but the last core position, which is mapped to its predecessor. $\square$

Proposition 2 (Synchronization). *Every automaton in the construction is synchronizing.*

Proof. By Lemma 3, first map Q into C . By Lemma 4, the semigroup restricted to C contains an adjacent idempotent merge. Conjugating γ by powers of the full cycle α moves this adjacent merge to every edge of the cycle. Applying these merges successively reduces C to one state. Prepending the entry word therefore yields a reset word for Q . $\square$

7 The full-set boundary

Suppose $S = Q$. A synchronizing automaton must contain a singular letter: if every letter were a permutation, every word would be a permutation and no reset word could exist. A singular letter compresses Q in one step, while the empty word is not successful. Hence

$$\tau_{\text{ac}}(\mathcal{A}; Q, A) = 1.$$

This proves the remaining part of Theorem 1.

8 Computational audit

The proof is analytic, but we also implemented an independent verifier. For each pair

$$1 \leq r \leq 30, \quad 3 \leq L \leq 40,$$

the program constructs the automaton, performs breadth-first search in the subset automaton to obtain the exact first-success distance, checks strong connectivity, and evaluates an explicit reset word built from the core entry and conjugated adjacent merges. This gives 1,140 tested parameter pairs, up to $n = 70$. Every case passed:

Check	Result
Parameter pairs	1,140
Exact distance = L	1,140 / 1,140
Strong connectivity	1,140 / 1,140
Explicit reset word	1,140 / 1,140
Failures	0

The verifier uses no external packages and is intended as reproducibility support rather than as a substitute for the proof.

9 Discussion

The second diagonal is the first infinite proper-subset line on which the avoid-or-compress upper bound can be tested against simultaneously binary, strong-connectivity, and synchronization constraints. Theorem 1 shows that none of these restrictions improves the general value: the full $n - |A|$ delay remains attainable.

This also isolates where more interesting behavior can first occur. On the third diagonal $|S| - |A| = 3$, the classical subset-compression bound already beats $n - |A|$ when $|S| = n - 1$, creating a boundary phase transition. That case is treated separately in the companion paper on the third diagonal.

Data and code availability

The accompanying script `N017_SECOND_DIAGONAL_VERIFY.py` reconstructs the family and reproduces the computational audit reported above.

References

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